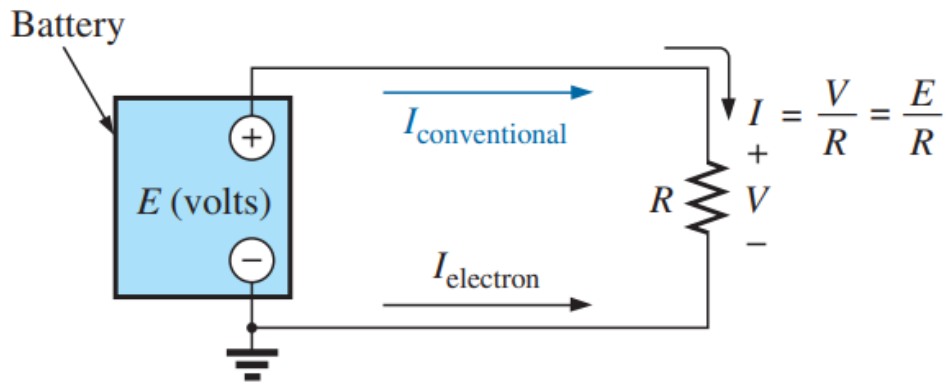
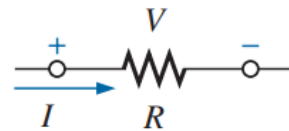
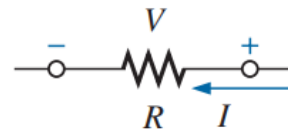




For any resistor, in any network, the direction of current through a resistor will define the polarity of the voltage drop across the resistor



(a)

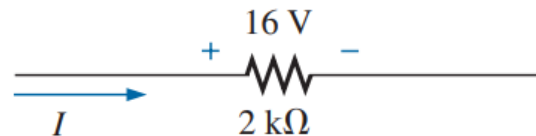


(b)

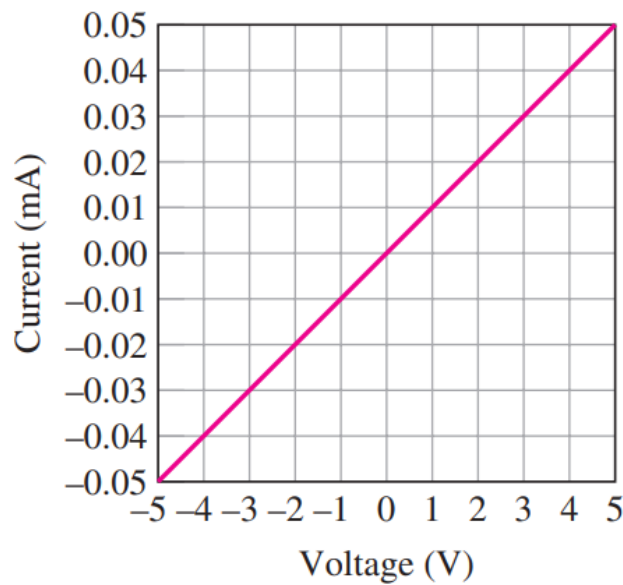
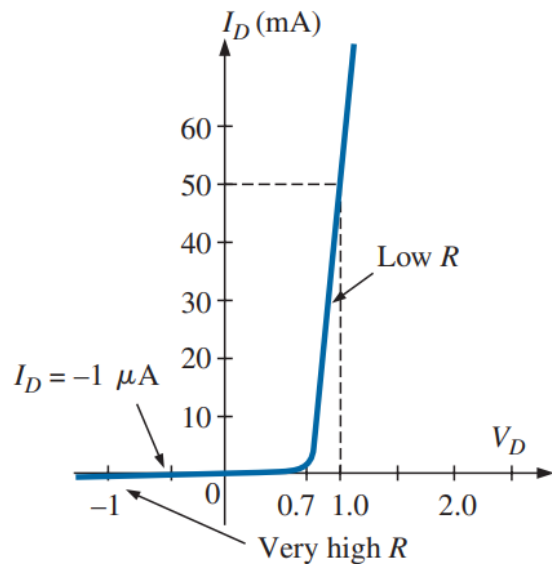
Power supplied=Power absorbed.

$Ei=Vi$ ,  $E=V=iR$

Voltage rise ( $E$ ) = Voltage drop( $V$ )



Determine the magnitude of the current flowing through a  $4.7 \text{ k}\Omega$  resistor if the voltage across it is (a)  $1 \text{ mV}$ ; (b)  $10 \text{ V}$ ; (c)  $4e^{-t} \text{ V}$ ; (d)  $100 \cos(5t) \text{ V}$ ; (e)  $-7 \text{ V}$ .



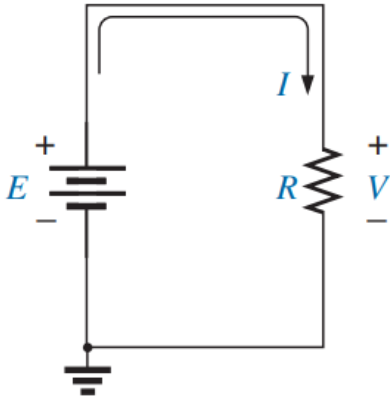
$$\frac{i}{v} = \frac{1}{R} = G$$

$$P = VI = V \frac{V}{R} = \frac{V^2}{R}$$

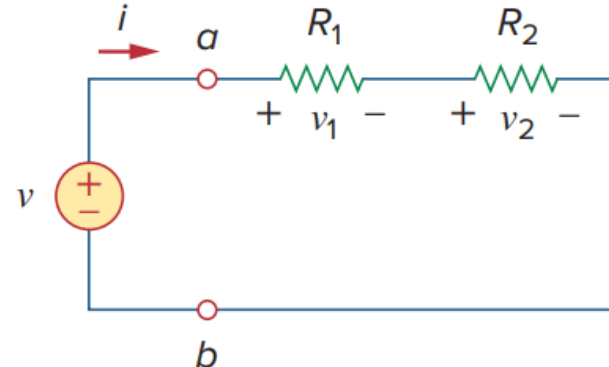
$$P = VI = IR \times I = I^2 R$$

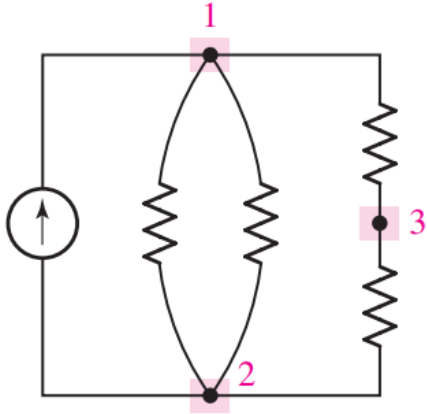
$$1 \text{ S} = 1 \text{ } \Omega^{-1} = 1 \text{ A/V}$$

$$p = vi = v^2 G = \frac{i^2}{G}$$



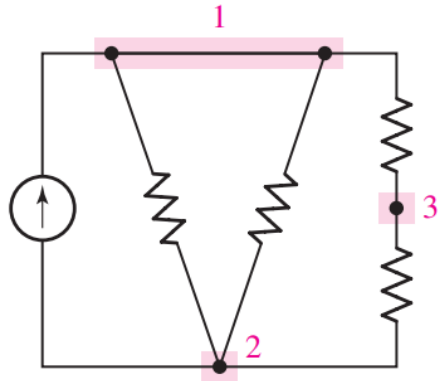
- For only two elements, Voltage source and a resistor, we can employ Ohm's law to determine current.
- We know voltage across the resistor, we know current!
- How can we determine voltage and current if there are more than one passive elements in the circuit connected in various arrangement?
- Circuits must be analyzed to determine a complete set of voltages and currents.
- In addition to Ohm's law, two more simple laws are needed.
- These new laws are **Kirchhoff's current law (KCL)** and **Kirchhoff's voltage law (KVL)**.





- A **node** in an electrical circuit is a point at which two or more circuit elements are joined together.
- Two nodes with no element in between are the same node.
- **Voltage value at any node or collection of same nodes is constant.**

If no node was encountered more than once as you move along the circuit, then the set of nodes and elements that we have passed through is defined as a **path**.

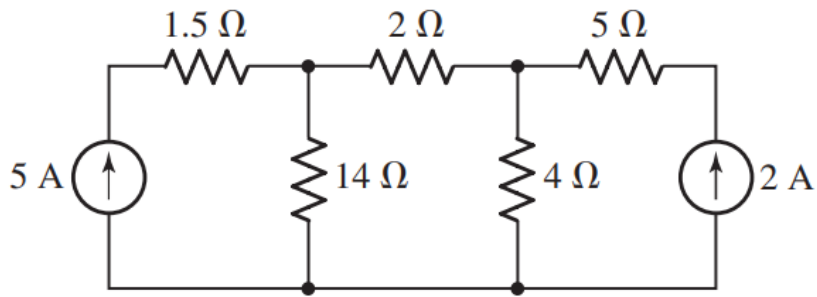


A **branch** represents a single element such as a voltage source or a resistor.

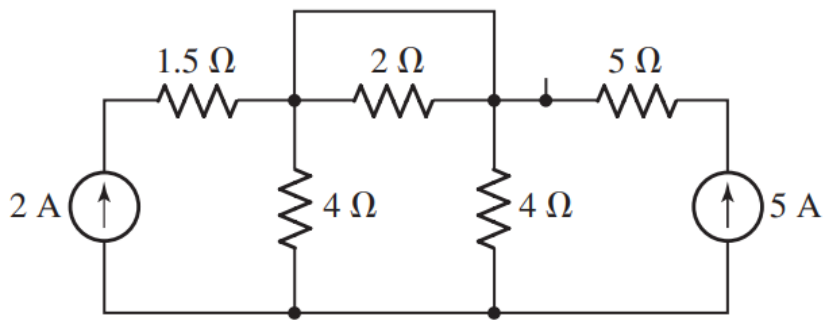
If the node at which we started is the same as the node on which we ended, then the path is, by definition, a closed path or a **loop**.

**Independent loop:** A loop is said to be independent if it contains at least one branch which is not a part of any other independent loop.

Referring to the circuit depicted in Fig. below, count the number of (a) nodes; (b) elements; (c) branches

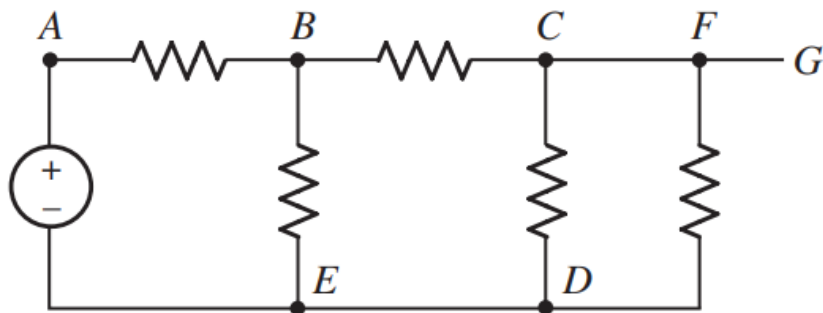


■ FIGURE 3.44



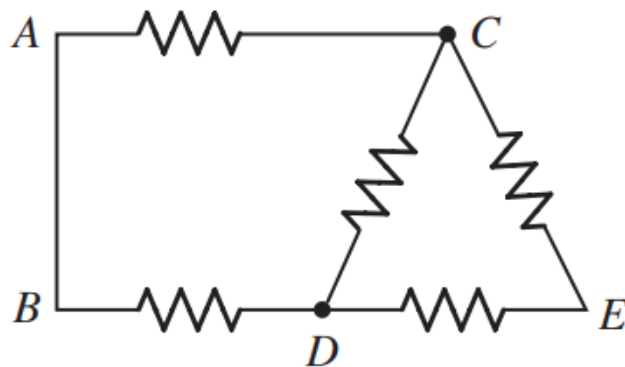
3. For the circuit of Fig. 3.46:

- (a) Count the number of nodes.
- (b) In moving from  $A$  to  $B$ , have we formed a path? Have we formed a loop?
- (c) In moving from  $C$  to  $F$  to  $G$ , have we formed a path? Have we formed a loop?



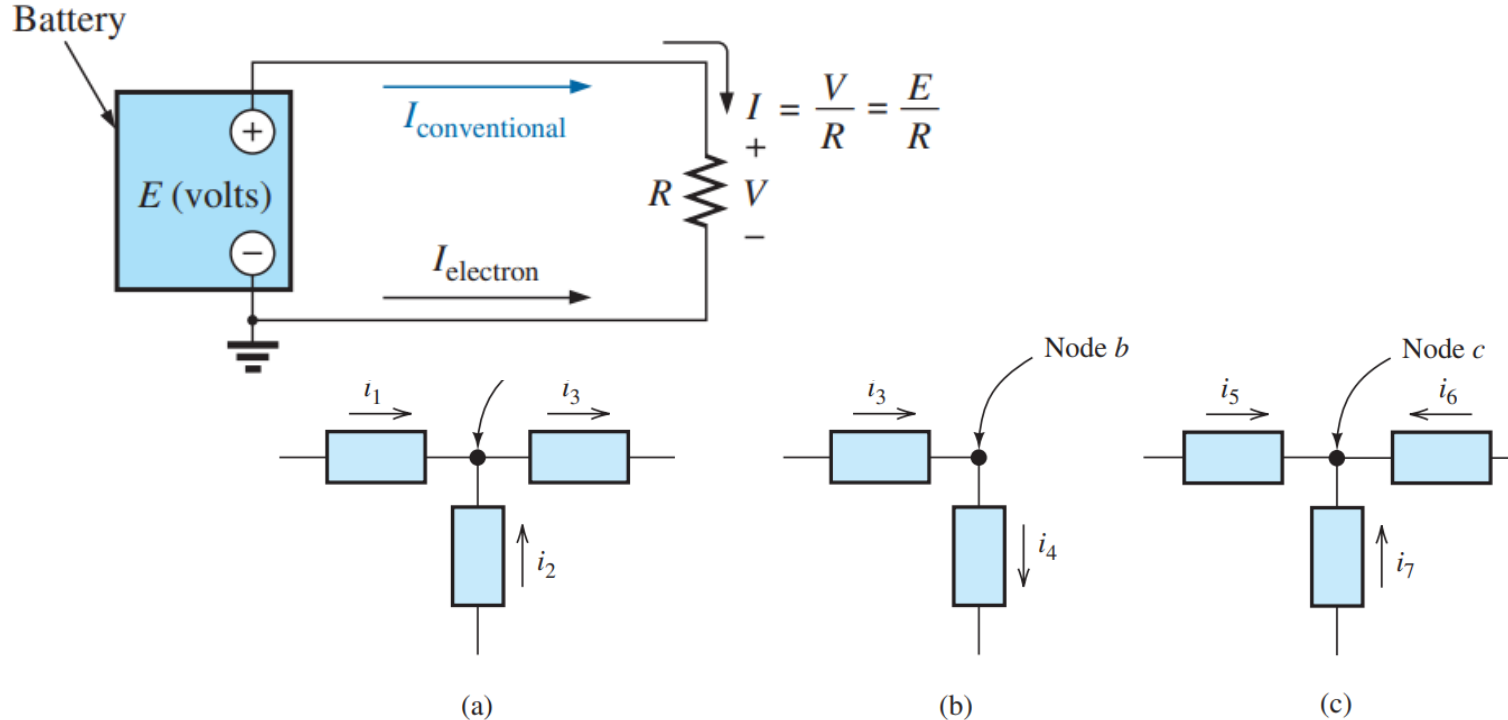
5. Refer to the circuit of Fig. 3.47, and answer the following:

- (a) How many distinct nodes are contained in the circuit?
- (b) How many elements are contained in the circuit?
- (c) How many branches does the circuit have?
- (d) Determine if each of the following represents a path, a loop, both, or neither:
  - (i)  $A$  to  $B$
  - (ii)  $B$  to  $D$  to  $C$  to  $E$
  - (iii)  $C$  to  $E$  to  $D$  to  $B$  to  $A$  to  $C$
  - (iv)  $C$  to  $D$  to  $B$  to  $A$  to  $C$  to  $E$

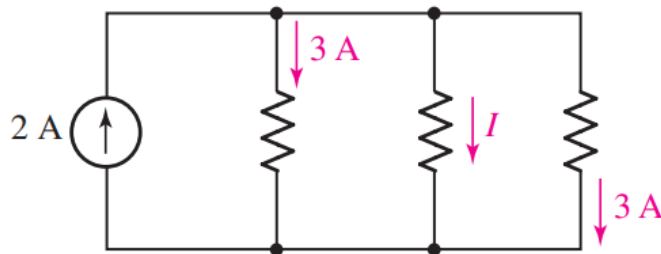
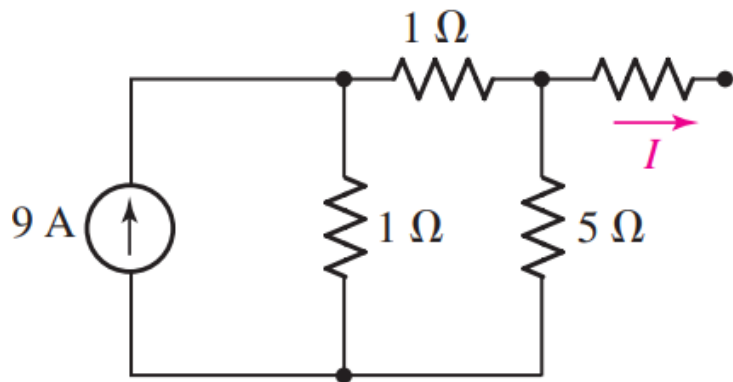


**KCL:** The algebraic sum of the currents entering any node is zero.

The sum of the currents entering a node equals the sum of the currents leaving a node.

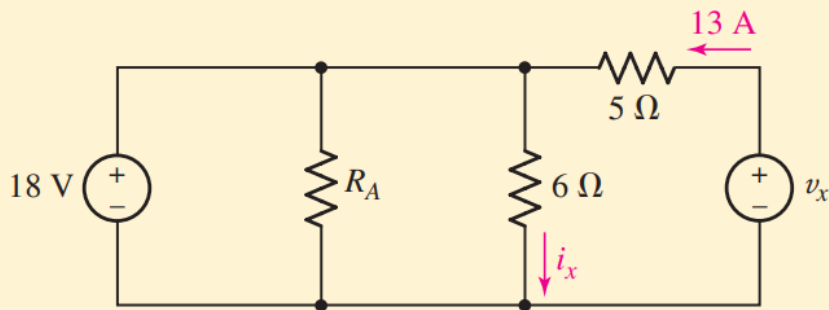


**Figure 1.18** Partial circuits showing one node each to illustrate Kirchhoff's current law.



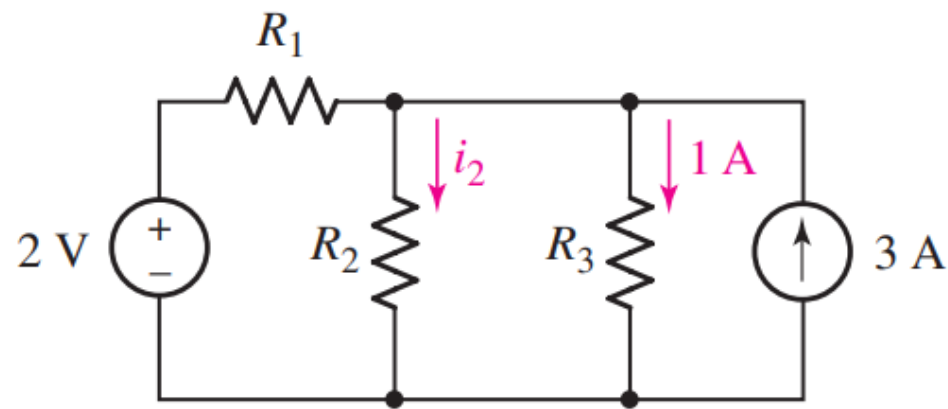
$i_{R_1}$

3.1 Count the number of branches and nodes in the circuit in Fig. 3.4. If  $i_x = 3$  A and the 18 V source delivers 8 A of current, what is the value of  $R_A$ ? (Hint: You need Ohm's law as well as KCL.)

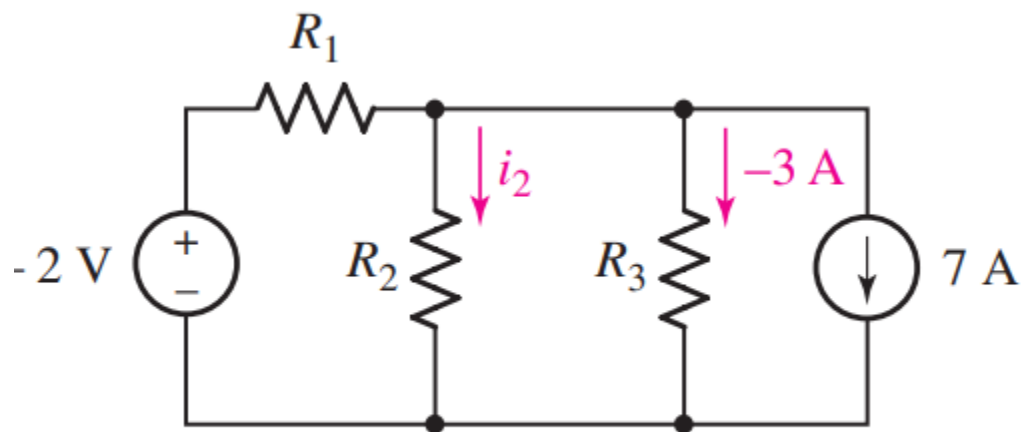


■ FIGURE 3.4

9. In the circuit shown in Fig. 3.51, the resistor values are unknown, but the 2 V source is known to be supplying a current of 7 A to the rest of the circuit. Calculate the current labeled  $i_2$ .

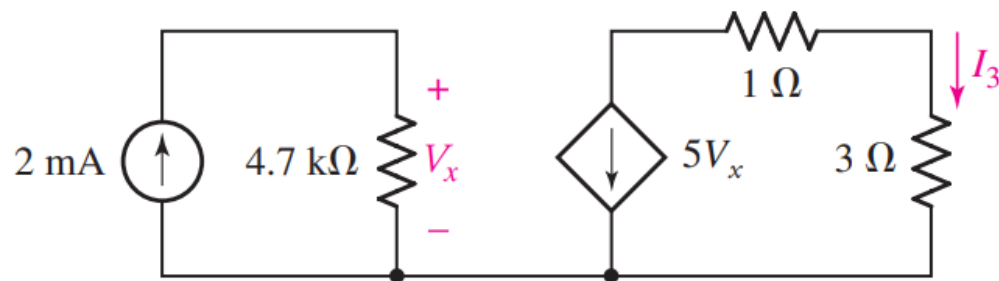


10. The voltage source in the circuit of Fig. 3.52 has a current of 1 A flowing out of its positive terminal into resistor  $R_1$ . Calculate the current labeled  $i_2$ .



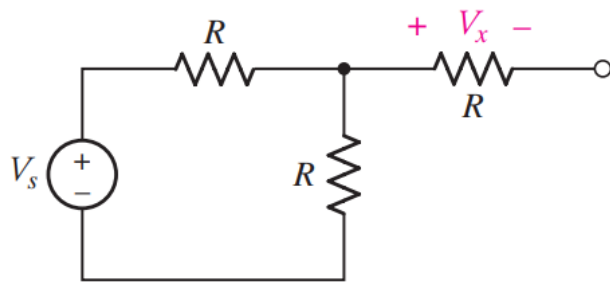
■ FIGURE 3.52

13. Determine the current labeled  $I_3$  in the circuit of Fig. 3.55.



■ FIGURE 3.55

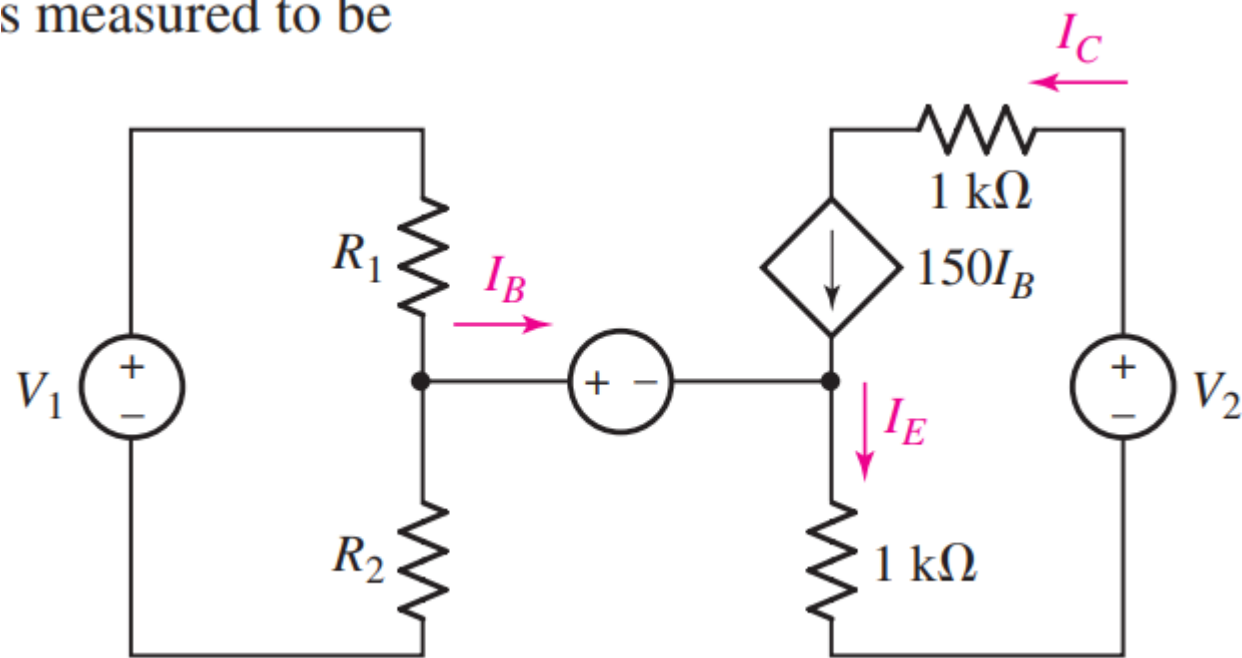
14. Study the circuit depicted in Fig. 3.56, and explain (in terms of KCL) why the voltage labeled  $V_x$  must be zero.

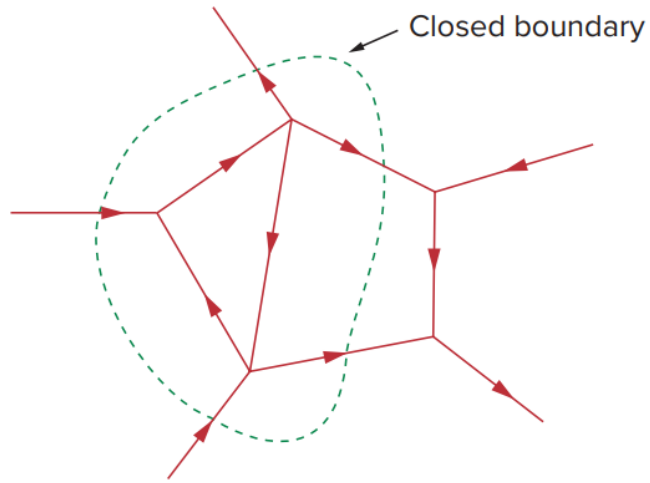


■ FIGURE 3.56

12. For the circuit of Fig. 3.54 (which is a model for the dc operation of a bipolar junction transistor biased in forward active region),  $I_B$  is measured to be  $100\ \mu\text{A}$ . Determine  $I_C$  and  $I_E$ .

s measured to be





**Figure 2.17**

Applying KCL to a closed boundary.

**total current entering the closed surface is equal to the total current leaving the surface.**